

Data Analysis with R and Python — Quiz 1 — April 7, 2025

NAME:

ROLL NUMBER:

For this quiz, we will use the UK 2024 (provisional) road safety vehicles data available from <https://www.data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-accidents-safety-data>. Use the following code to read the data in R from a copy of the data at the course website.

```
DATALOC <- "https://deepayan.github.io/BSDS/2025-01-DARP/slides/data/rcs2024.csv"
rcs2024 <- read.csv(DATALOC)
```

Each row in the dataset describes a vehicle involved in a collision. Multiple vehicles may be involved in a single collision. The `collision_index` variable is an identifier for the collision a particular vehicle was involved in.

The `sex_of_driver` variable is coded as an integer, with values 1 = Male, 2 = Female, 3 = Not known. Missing values are coded as `-1`.

Question 1: Identify the specific collision that involved the largest number of vehicles (hint: the number of vehicles involved was 25). For this collision, [5]

- How many of the drivers were Male?
- How many of the drivers were Female?
- Ignoring missing values, what was the average age of the Male drivers?
- Ignoring missing values, what was the average age of the Female drivers?

Question 2: Obtain the subset of the data for collisions that involved exactly one vehicle.

[10]

- How many such collisions are recorded?
- For what proportion of such collisions is the age of the driver missing?
- For what proportion of such collisions is the sex of the driver Not known?
- Ignoring missing values, what was the average age of the Male drivers in such collisions?
- Ignoring missing values, What was the average age of the Female drivers in such collisions?
- Ignoring missing values, What was the average age of drivers with sex Not Known in such collisions?
- What is the most frequent vehicle type involved in such collisions? The meaning of the integer codes of the `vehicle_type` variable is explained in the attached form.
- What is the *least* frequent vehicle type involved in such collisions? Exclude all vehicle types that were not involved in any recorded accident.

Question 3: Using graphical tools you have learned about, study the empirical distribution of `age_of_driver` for different values of `sex_of_driver` (in the entire dataset). Briefly summarize your observations, keeping the following underlying questions in mind: [10]

- Does the age distribution of drivers involved in accidents vary by their sex?
- If we wanted to summarize the age distributions by sex, what would be an appropriate measure (e.g., sample mean, sample median, something else?)

Good luck!

2.26 VEHICLE REGISTRATION MARK												
Vehicle 001												
Vehicle 002												
Vehicle 003												
Vehicle 004												

2.35 WAS THE VEHICLE LEFT HAND DRIVE ✗		VEHICLE			
		1	2	3	4
No	1				
Yes	2				

2.5 / 2.5a TYPE OF VEHICLE ✗					
Car	09				
Taxi / Private hire car	08				
Van - Goods vehicle 3.5 tonnes mgw and under	19				
Goods vehicle over 3.5 tonnes mgw and under 7.5 tonnes mgw	20				
Goods vehicle 7.5 tonnes mgw & over	21				
Goods vehicle - unknown weight	98				
M/cycle 50cc and under	02				
M/cycle over 50cc and up to 125cc	03				
M/cycle over 125cc and up to 500cc	04				
Motorcycle over 500cc	05				
Motorcycle - cc unknown	97				
Electric Motorcycle	23				
Pedal cycle	01				
Bus or coach (17 or more passenger seats)	11				
Minibus (8-16 passenger seats)	10				
Agricultural vehicle (include diggers etc)	17				
Ridden horse	16				
Mobility scooter	22				
Tram / Light rail	18				
Other 1	90				
vehicle 2	90				
3	90				
4	90				

2.6 TOWING AND ARTICULATION ✗					
No tow or articulation	0				
Articulated vehicle	1				
Double or multiple trailer	2				
Caravan	3				
Single trailer	4				
Other tow	5				

2.22 AGE OF DRIVER (Estimate if necessary)									
Vehicle 001					Vehicle 002				
Vehicle 003					Vehicle 004				

2.27 DRIVER HOME POSTCODE or Code: 1- Unknown 2- Non UK Resident 3 - Parked & unattended												
Vehicle 001												
Vehicle 002												
Vehicle 003												
Vehicle 004												

2.23 BREATH TEST ✗		VEHICLE			
		1	2	3	4
Not applicable	0				
Positive	1				
Negative	2				
Not requested	3				
Refused to provide	4				
Driver not contacted at time of col'	5				
Not provided (medical reasons)	6				

2.24 HIT AND RUN ✗					
Not hit and run	0				
Hit and run	1				
Non-stop vehicle, not hit	2				

2.21 SEX OF DRIVER ✗					
Male	1				
Female	2				
Not known	3				

2.9 VEHICLE LOCATION AT TIME OF ACCIDENT RESTRICTED LANE/ AWAY FROM MAIN C'WAY ✗					
On main carriageway not in restricted lane	00				
Tram / Light rail track	01				
Bus lane	02				
Busway (inc. guided busway)	03				
Cycle lane (on main carriageway)	04				
Cycleway or shared use footway (not part of main carriageway)	05				
On lay-by / hard shoulder	06				
Entering lay-by/ hard shoulder	07				
Leaving lay-by / hard shoulder	08				
Footway (pavement)	09				

2.10 JUNCTION LOCATION OF VEHICLE ✗					
Not at or within 20m of junction	0				
Approaching junction or waiting / parked at junction approach	1				
Cleared junction or waiting/ parked at junction exit	2				
Leaving roundabout	3				
Entering roundabout	4				
Leaving main road	5				
Entering main road	6				
Entering from slip road	7				
Mid junction- on roundabout or on main road	8				

2.7 MANOEUVRES ✗					
Reversing	01				
Parked	02				
Waiting to go ahead but held up	03				
Slowing or stopping	04				
Moving off	05				
U turn	06				
Turning left	07				
Waiting to turn left	08				
Turning right	09				
Waiting to turn right	10				
Changing lane to left	11				
Changing lane to right	12				
O'taking moving veh on its offside	13				
O'taking stationary veh on its offside	14				
Overtaking on nearside	15				
Going ahead left hand bend	16				
Going ahead right hand bend	17				
Going ahead other	18				

2.11 SKIDDING AND OVERTURNING ✗		VEHICLE			
		1	2	3	4
No skidding, jack-knifing or overturning	0				
Skidded	1				
Skidded and overturned	2				
Jack - knifed	3				
Jack - knifed and overturned	4				
Overturned	5				

2.12 HIT OBJECT IN CARRIAGEWAY ✗					
None	00				
Previous accident	01				
Roadworks	02				
Parked vehicle	04				
Bridge - roof	05				
Bridge - side	06				
Bollard / Refuge	07				
Open door of vehicle	08				
Central island of roundabout	09				
Kerb	10				
Any animal (except ridden horse)	12				
Other object	11				

2.13 VEHICLE LEAVING CARRIAGEWAY ✗					
Did not leave carriageway	0				
Left carriageway nearside	1				
Left carriageway nearside and rebounded	2				
Left carriageway straight ahead at junction	3				
Left carriageway offside onto central reservation	4				
Left carriageway offside onto central reserve and rebounded	5				
Left carriageway offside and crossed central reservation	6				
Left carriageway offside	7				
Left carriageway offside and rebounded	8				

2.14 FIRST OBJECT HIT OFF CARRIAGEWAY ✗					
None	00				
Road sign / Traffic signal	01				
Lamp post	02				
Telegraph pole / Electricity pole	03				
Tree	04				
Bus stop / Bus shelter	05				
Central crash barrier	06				
Nearside or offside crash barrier	07				
Submerged in water (completely)	08				
Entered ditch	09				
Wall or fence	11				
Other permanent object	10				

2.16 FIRST POINT OF IMPACT ✗					
Did not impact	0				
Front	1				
Back	2				
Offside	3				
Nearside	4				

2.29 JOURNEY PURPOSE OF DRIVER/RIDER ✗					
Journey as part of work	1				
Commuting to / from work	2				
Taking school pupil to/from school	3				
Pupil riding to / from school	4				
Other	5				
Not known	6				

Subject to local directions, boxes with a grey background need not be completed if already recorded